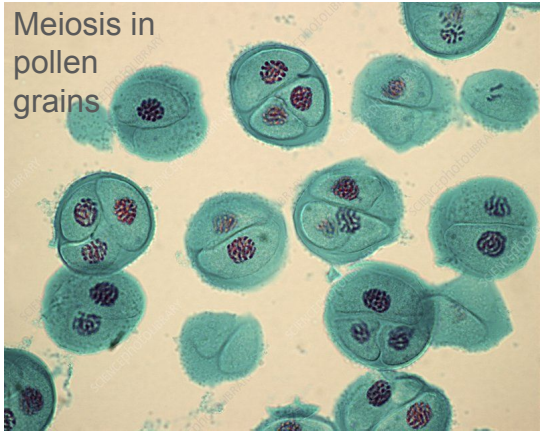
BME 105: Genetics in the Genomics Era

Lecture 6: Chromosomes, ploidy, mitosis, meiosis



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Assistant Professor

Course website on canvas.ucsc.edu

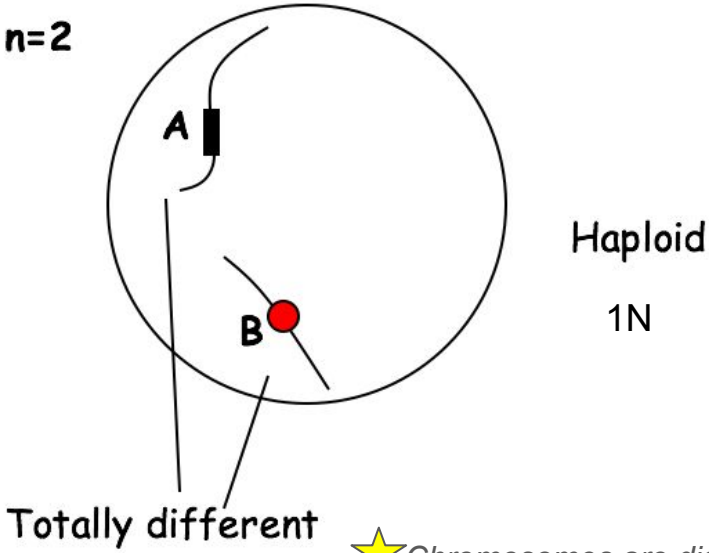
Lecture 6 Learning Objectives

- Understand the difference between chromosome number and ploidy.
- Understand how chromosomes make more of themselves through mitosis
  - Know the stages and cellular chromosomal contents of each step.
- Understand how chromosomes are packaged into gametes through meiosis, to be inherited by the next generation
  - Know the stages and cellular chromosomal contents: When do homologs segregate? When do sister chromatids separate?
- Know where mitosis and meiosis occur in humans ~ the difference between germline and soma
- Be able to describe fertilization in chromosomal terms
- Understand what sister chromatids are, when they form, and when they separate in mitosis and meiosis.
- Know the stages of the cell cycle - know where mitosis fits into the cell cycle.
- Be able to articulate how genetic variation is created during meiosis (part 1): by random assortment of homologs on the meiosis I metaphase plate
  - In lecture 8 we'll cover part 2: homologous chromosome crossing over and recombination during meiotic prophase I.

Chromosome number

Chromosome number = Autosome + sex chromosome

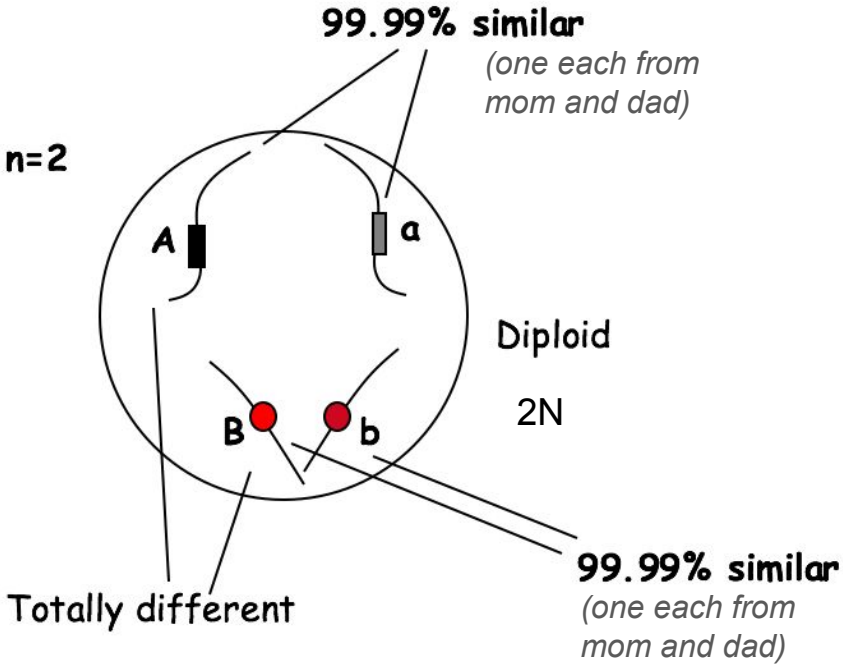
= n, haploid cell number/number of <u>non-homologous</u> chromosomes in a haploid cell, i.e., a gamete
(humans have 23)



Chromosomes are distinct functional and evolutionary entities

Ploidy = N, number of copies of <u>homologous</u> chromosomes in a cell

Haploids have 1N DNA content
Diploids have 2N DNA content (humans are diploids)
Tetraploids have 4N DNA content



Nonhomologous chromosome number

Species	**Chromosome number in haploid cells (n)**
Human	23
Monkey	21
Mouse	20
Fruit fly	4
C. Elegans (worm)	6
Corn	10

All of the above are diploid organisms, meaning **somatic cells have two copies of each chromosome**

(homologous)

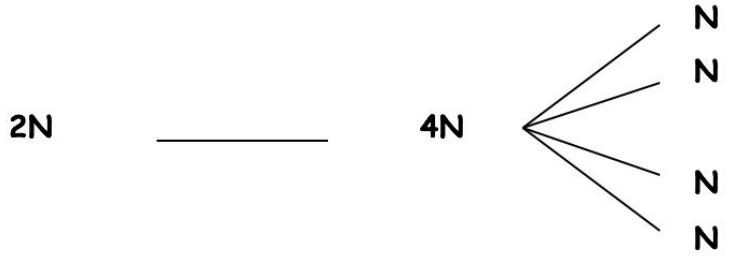
Mitosis vs Meiosis

Mitosis: a form of cell division that produces two daughter cells of identical genotypes

Used to make identical copies of DNA during normal cell division

Meiosis: in a diploid cell, it is a form of cell division that produces four haploid cells (gametes). Meiosis only occurs in specialized cells called germ cells.

(soma)



(germline)

Why did two replication-transmission mechanisms evolve?

Soma and germline cells have distinct chromosomal requirements for their division products (i.e., daughter cells)

Soma requirements:

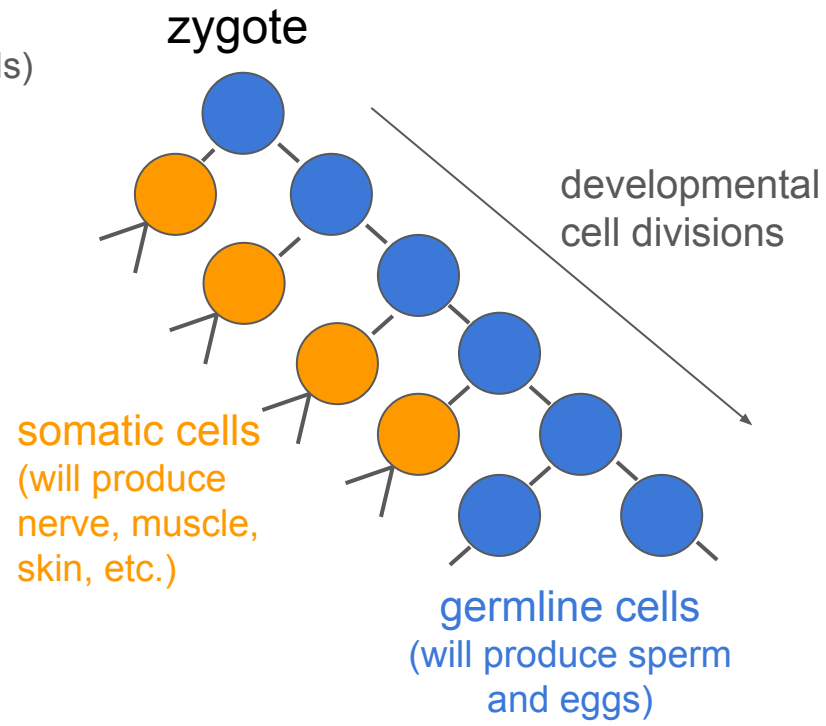
Faithful division

Retention of 2N (diploidy)

Germline requirements:

Proper pairing and segregation of homologs

Reduction of ploidy to 1N (haploidy)



Why did two replication-transmission mechanisms evolve?

Soma and germline cells have distinct chromosomal requirements for their division products (i.e., daughter cells)

mitosis

Soma requirements:

Faithful division

Retention of 2N (diploidy)

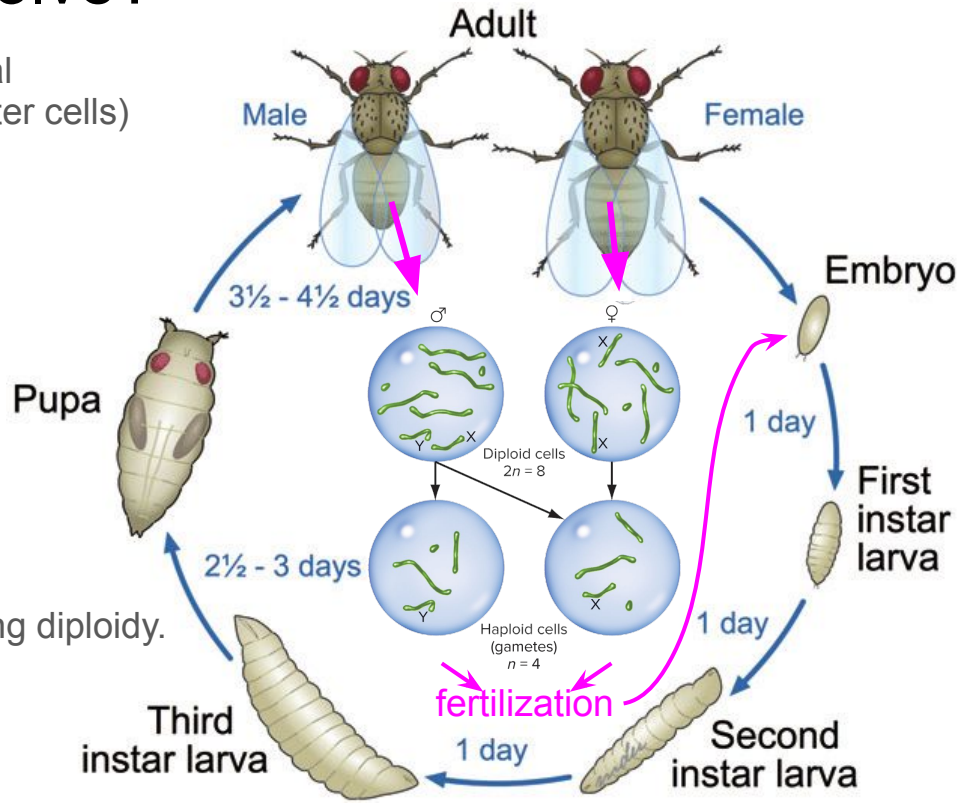
meiosis

Germline requirements:

Proper pairing and segregation of homologs

Reduction of ploidy to 1N (haploidy)

Fertilization unites parental chromosomes, restoring diploidy.



Cell cycle

The cell cycle occurs in somatic cells of the body

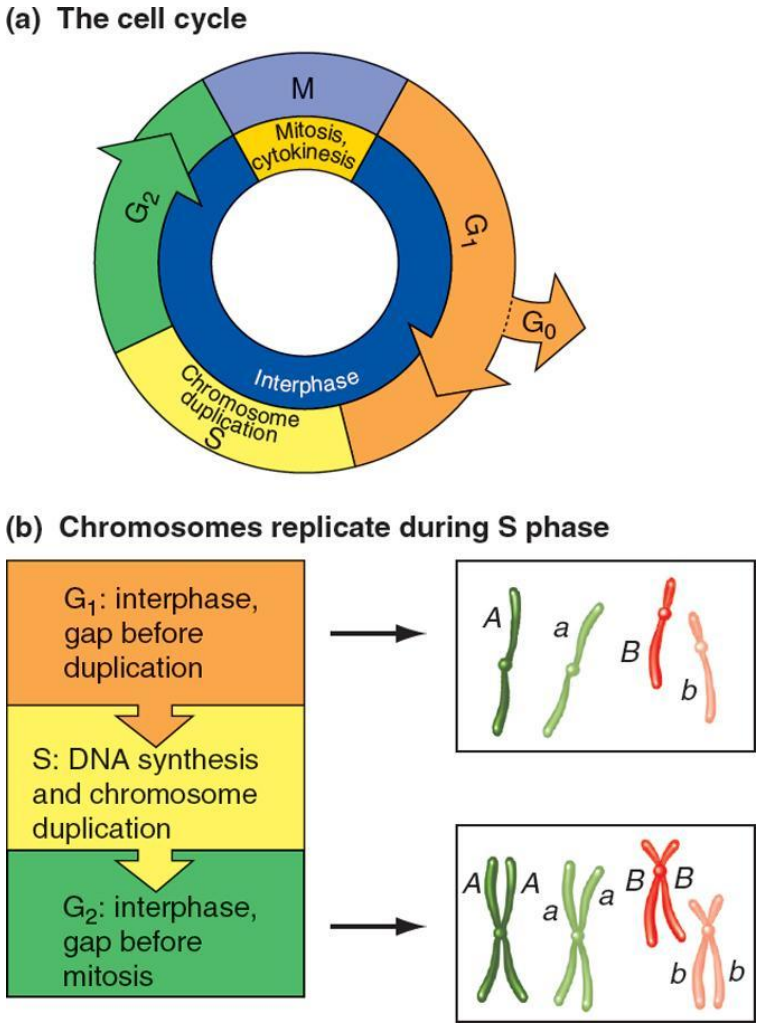
Cycles through:

1. Cell growth (G1 phase)
2. Replication of each chromosome (S phase),
3. Checkpoints before mitosis (G2 phase)
4. Segregation of replicated chromosomes into two daughter nuclei (M phase = mitosis)

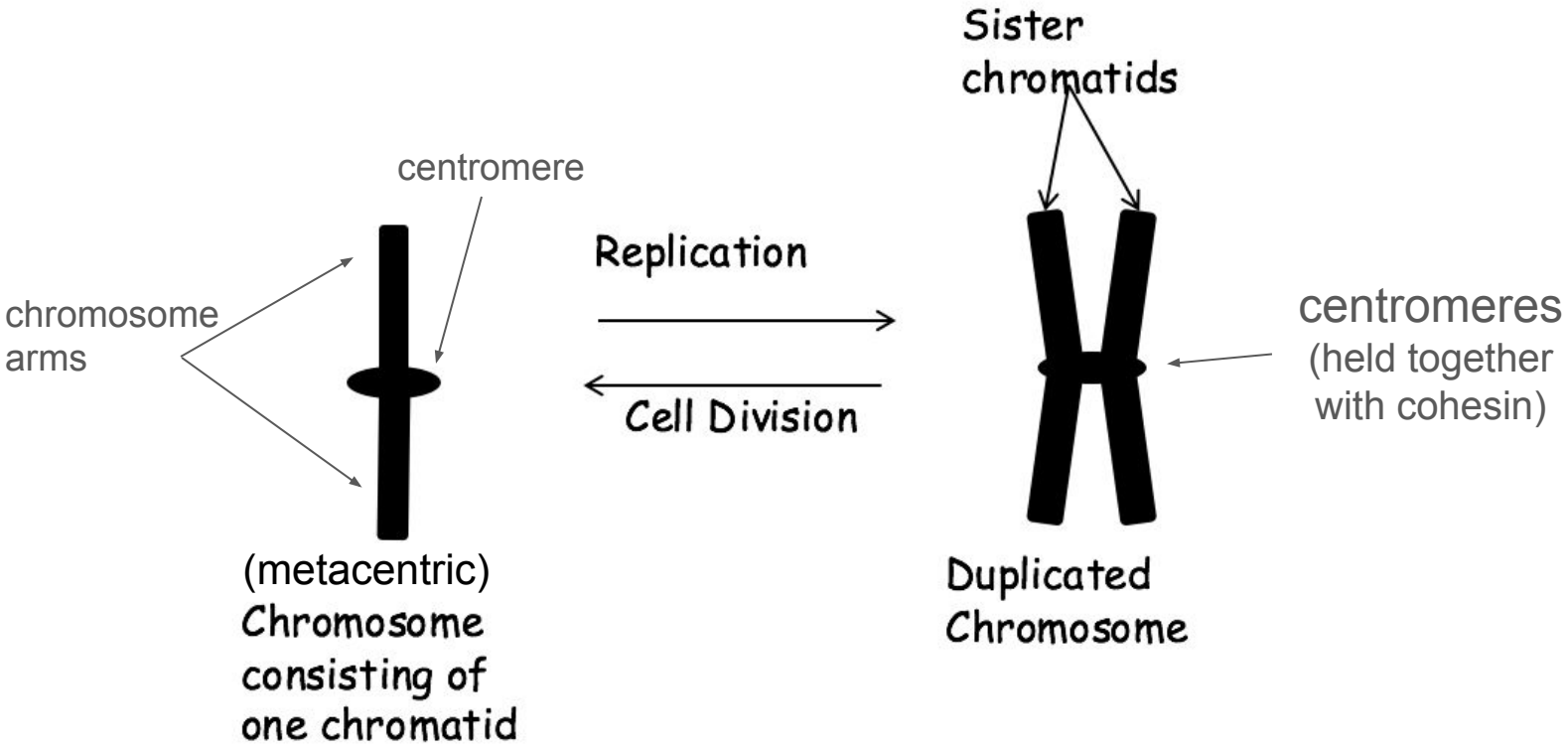
Gap phases = G1, G2 = growth and checkpoints

Interphase = G1, S, and G2 (i.e., gap and replication phases)

Exit: G0 - quiescence or resting phase. Not dividing



Sister chromatids



Sister chromatids occur after S phase

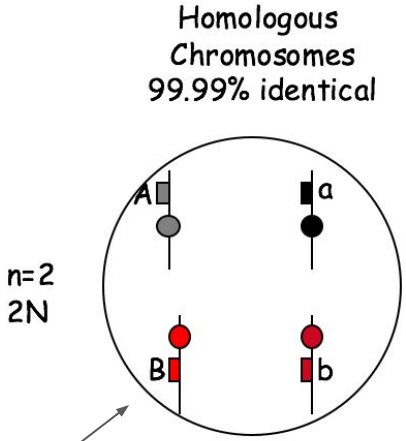
Each DNA molecule is a chromatid

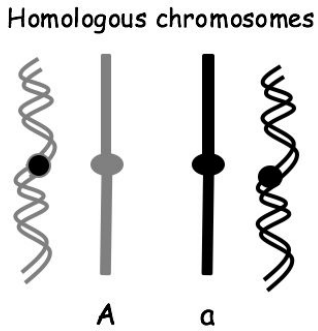
Two chromatids attached at centromere (via proteins) are sister chromatids

Homologous chromosomes are ~99.99% identical to each other.

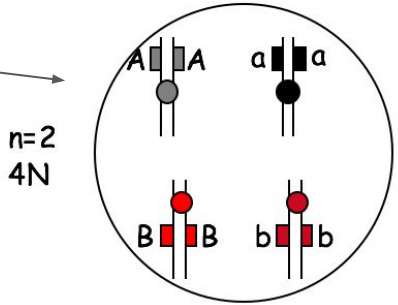
Sister chromatids are 100% identical to each other.

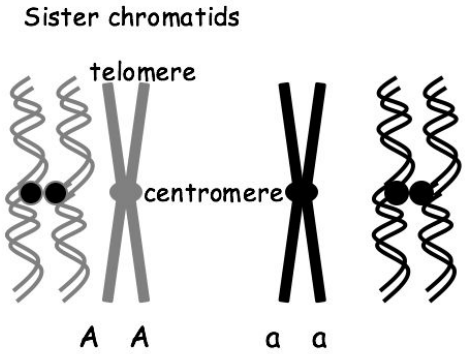
n chromosomes
N copies



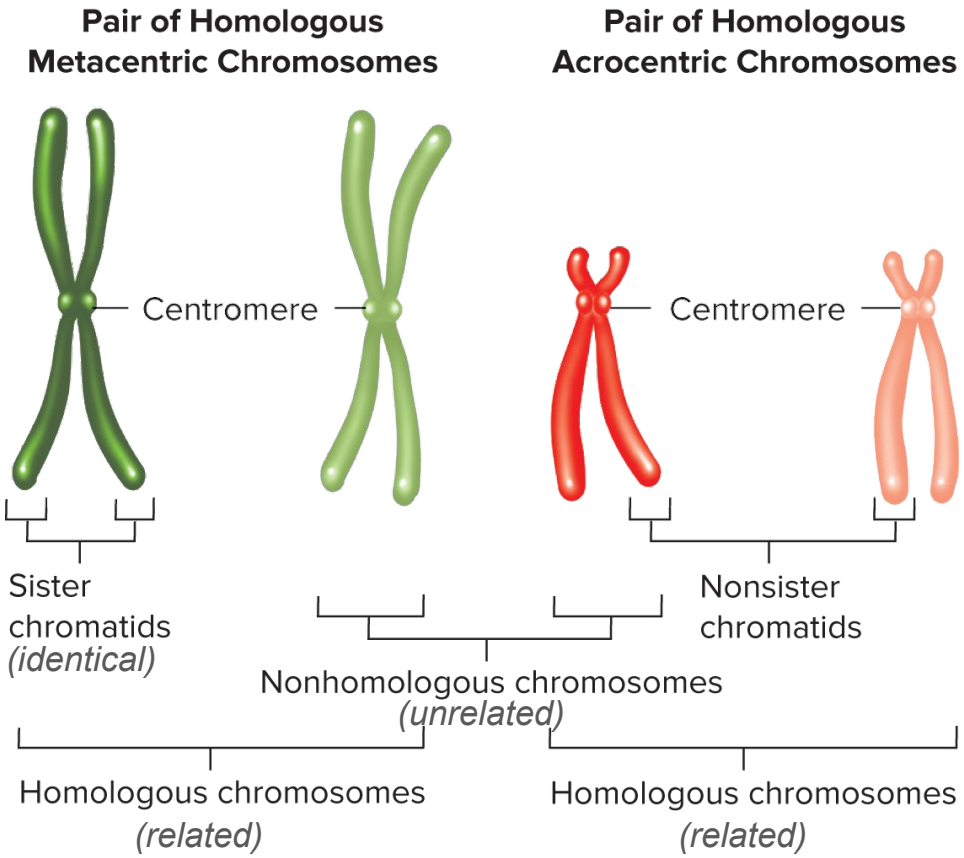


Replication of DNA





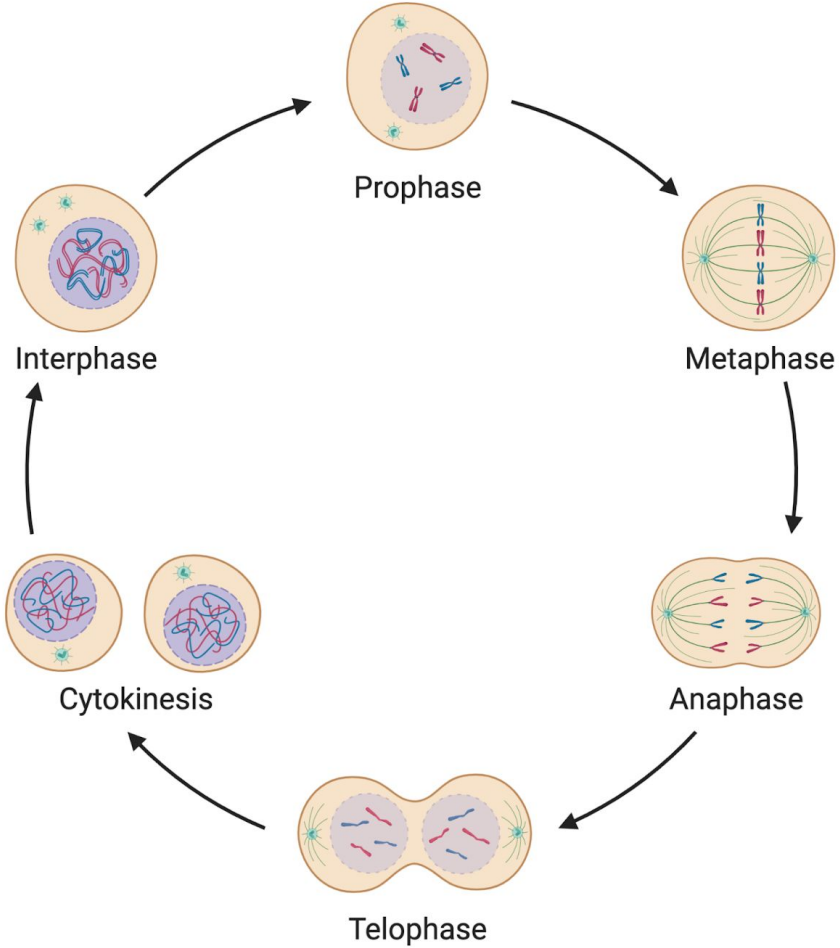
Sister chromatids vs. homologous chromosomes vs. non-homologous chromosomes



Mitosis

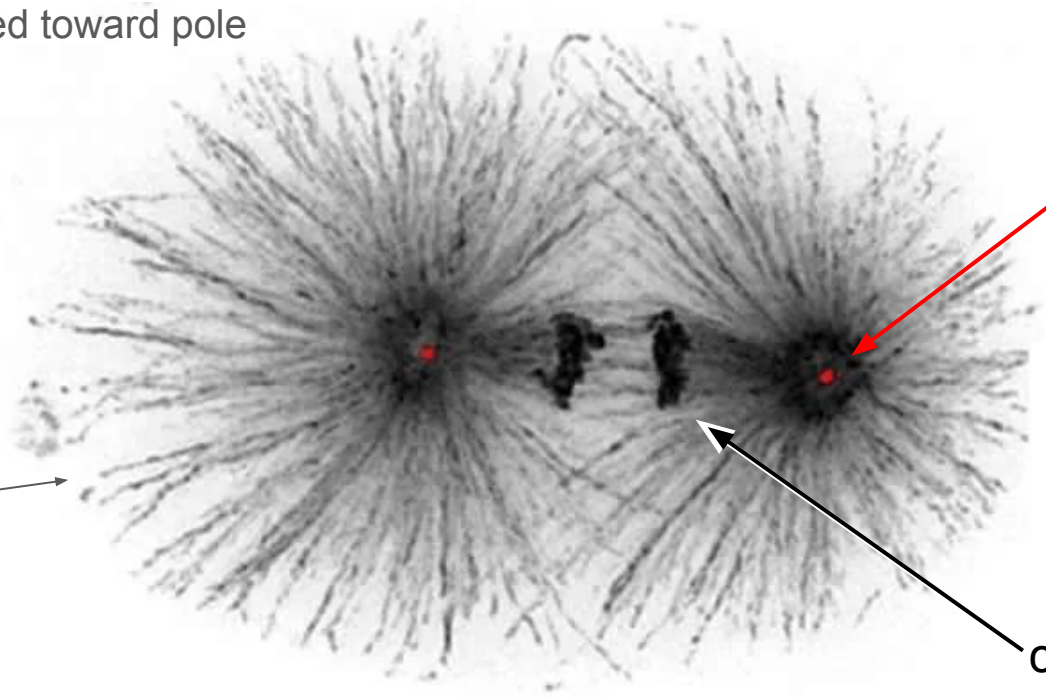
Stages of mitosis:

- Interphase: DNA is replicated
- Prophase: Chromosome condense
- Prometaphase: The spindle forms
- Metaphase: Chromosomes align at the cell's equator
- Anaphase: Sister chromatids move to opposite spindle poles
- Telophase: **Identical** sets of chromosomes are enclosed in two nuclei
- Cytokinesis: The cytoplasm divides



The mitotic spindle pulls nuclei to either pole and pushes the poles apart

Anaphase:
- chromosomes pushed toward pole
- poles pushed apart



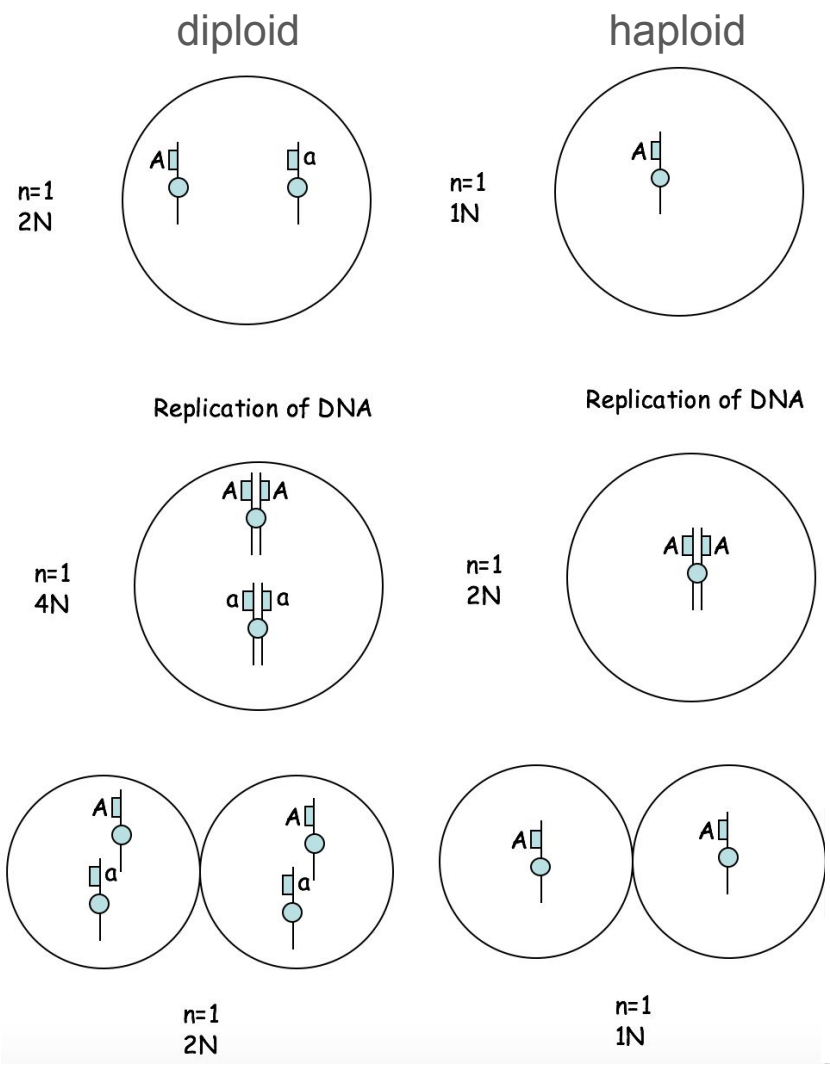
Cabral et al. 2019 Developmental Cell

Mitosis in diploid and haploid cells

Many organisms are haploid (N=1)

e.g., S.cerevisiae (baker's yeast)

Note: Bacteria and archaea are haploid, but divide by binary fission, not mitosis

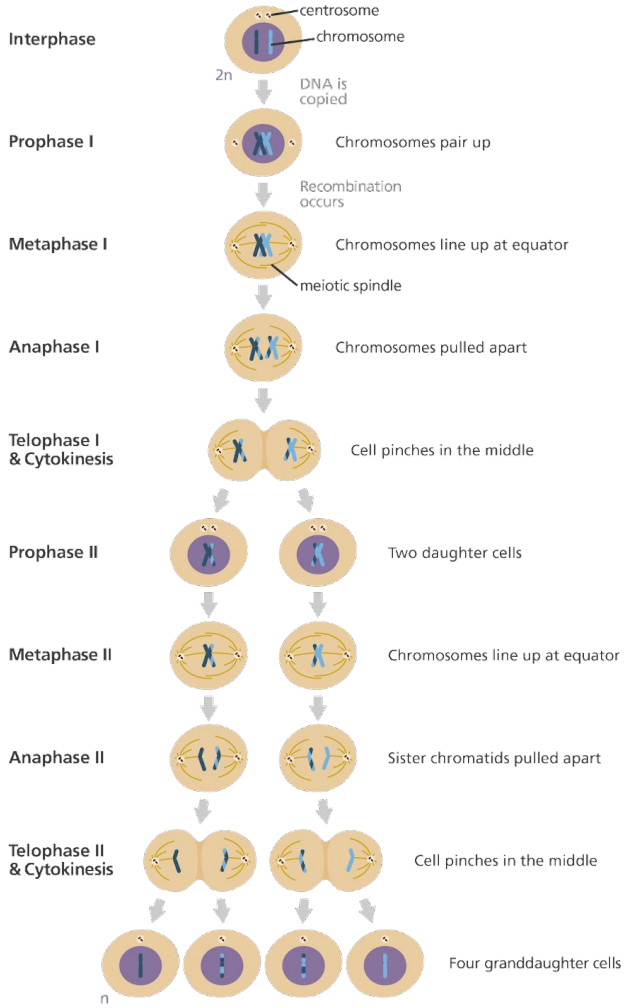


Stages of meiosis

Meiosis I: **Homologs** pair, exchange parts (recombination, see lecture 8), then **segregate**

Meiosis II: **Sister chromatids separate** to produce haploid gametes

*** Meiosis is the biological mechanism underlying Mendel's laws of independent segregation and assortment*



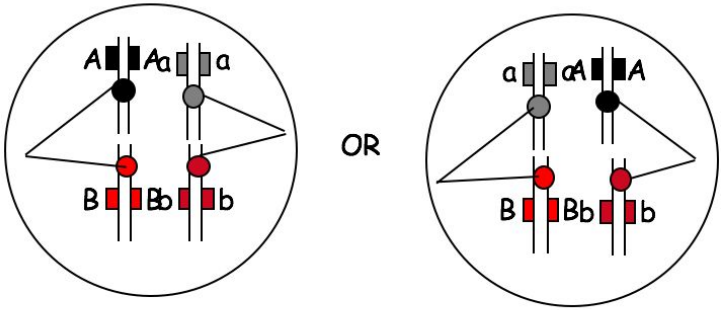
Random assortment

Meiosis I determines homolog assortment

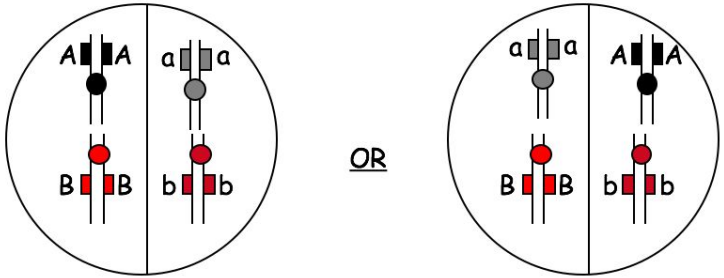
** Only half of the possible combinations realized in each meiosis **

At the end of meiosis I: sisters are still paired, but homologs are separated into different cells

Metaphase I



Anaphase I.
Centromeres do not separate
The two sister chromatids go to the same pole



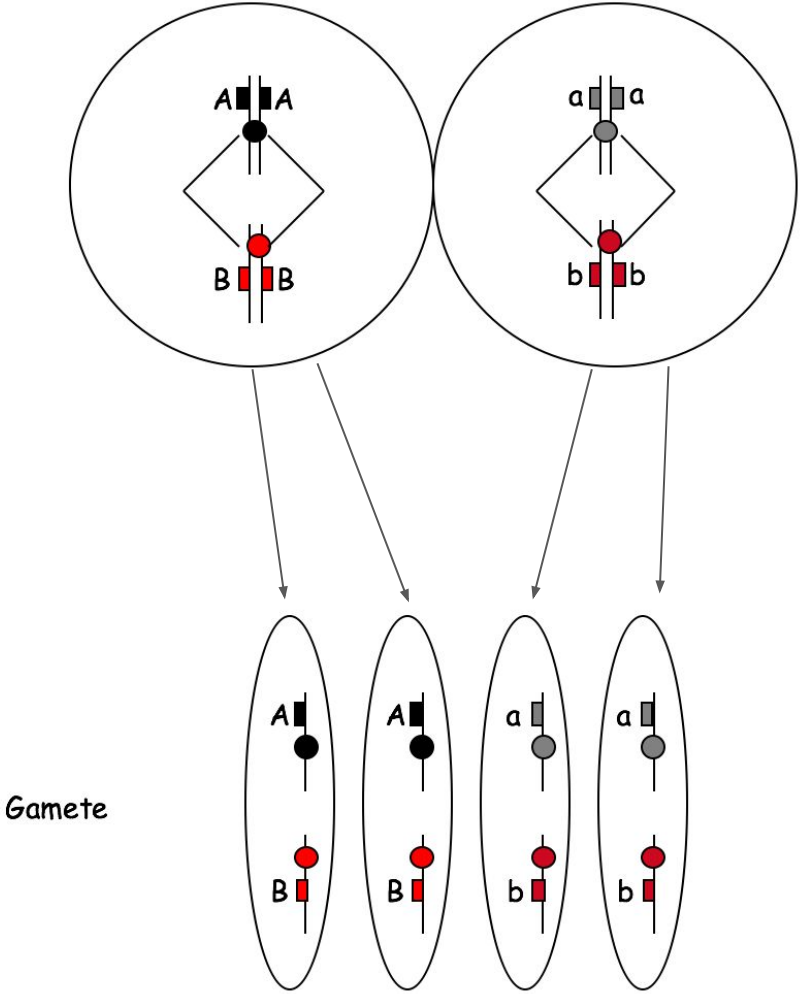
Telophase I and Cytokinesis

Metaphase II

The reduced number of chromosomes in each of the two cells align on the metaphase plate, divide to produce four haploid cells.

Cell division without intervening replication!!

Similar to mitotic metaphase in that each chromosome lines up at metaphase plate

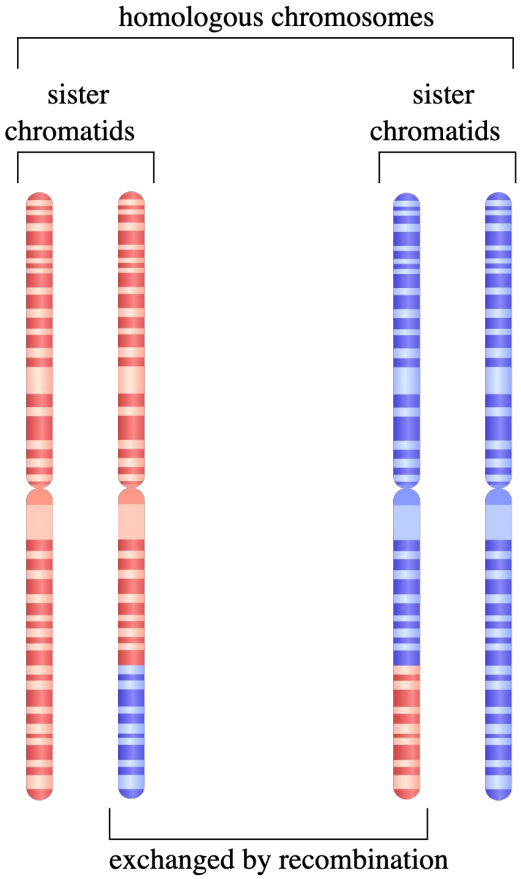


How does meiosis contribute to genetic diversity?

Independent assortment of different chromosomes in Metaphase I

Prior to this in meiosis I… homologous chromosomes pair and crossover, inducing recombination.

> Meiotic recombination generates **new allele combinations**

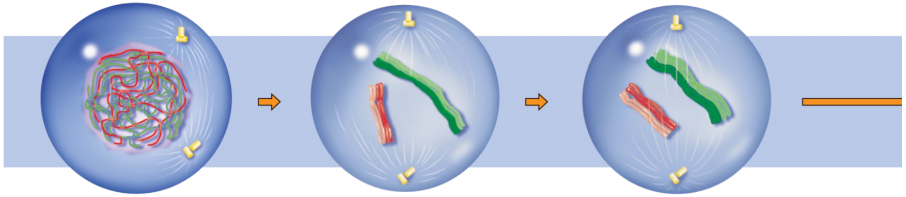


Emw, Wikipedia

Homologous chromosomes don’t only pair - they form a complex, crossover, and exchange sequence (i.e., recombine)

Recombination occurs during prophase I:

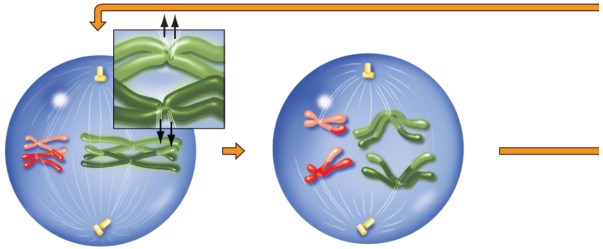
Meiosis I: A reductional division



Prophase I: Leptotene
1. Chromosomes thicken and become visible, but the chromatids remain invisible.
2. Centrosomes begin to move toward opposite poles.

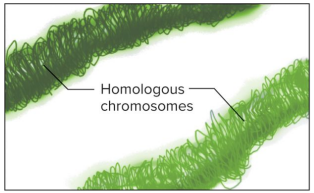
Prophase I: Zygotene
1. Homologous chromosomes enter *synapsis*.
2. The *synaptonemal complex* forms.

Prophase I: Pachytene
1. Synapsis is complete.
2. *Crossing-over*, genetic exchange between nonsister chromatids of a homologous pair, occurs.

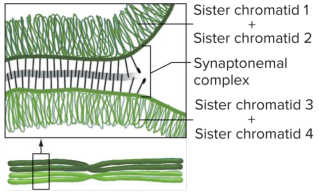


Metaphase I
1. Tetrads line up along the *metaphase plate*.
2. Each chromosome of a homologous pair attaches to fibers from opposite poles.
3. Sister chromatids attach to fibers from the same pole.

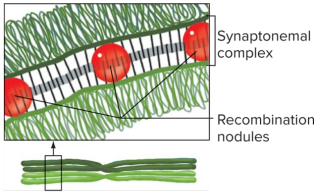
Anaphase I
1. Sister centromeres remain connected to each other.
2. The chiasmata dissolve.
3. Homologous chromosomes move to opposite poles.



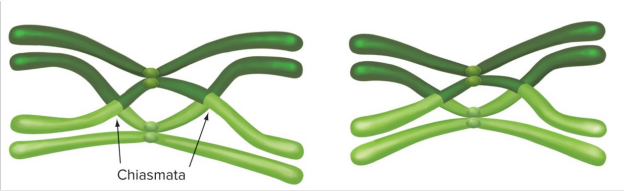
(a) Leptotene: Threadlike chromosomes begin to condense and thicken, becoming visible as discrete structures. Although the chromosomes have duplicated, the sister chromatids of each chromosome are not yet visible in the microscope.



(b) Zygotene: Chromosomes are clearly visible and begin pairing with homologous chromosomes along the synaptonemal complex to form a bivalent, or tetrad.



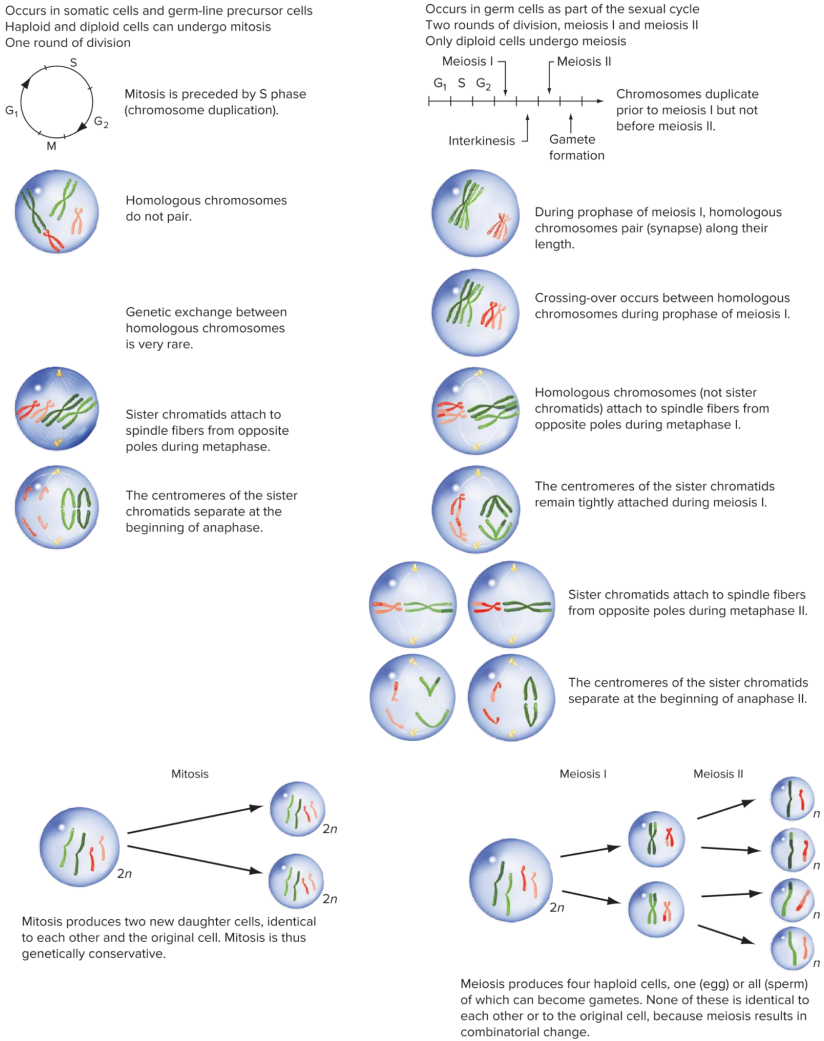
(c) Pachytene: Full synapsis of homologs. Recombination nodules appear along the synaptonemal complex.



(d) Diplotene: Bivalent pulls apart slightly, but homologous chromosomes remain connected due to recombination at crossover sites (chiasmata).

(e) Diakinesis: Further condensation of the bivalent.

Summary:
Mitosis vs meiosis

- Mitosis does not reduce ploidy, meiosis does
  - Mitosis is a reliable mechanism of somatic cell division

- Meiotic ploidy reduction is required for gamete formation in the germline
  - Chromosomes first divide
  - Then, homologs pair and segregate
    - Recombination occurs during pairing, in prophase I
  - Finally, sisters separate